

The strong Zeno product question in arXiv:0907.1199 was answered by arXiv:2012.15061

Literature-status note

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Status: literature already answered. The later Theorem 1.1 proves the exact strong-operator Zeno limit under the source hypothesis, and uniformly on bounded time intervals.

1 The source question

Let $B \geq 0$ be self-adjoint on a separable Hilbert space $\mathcal{H}$, let P be the orthogonal projection onto a closed subspace $\mathfrak{h}$, and assume

$$\text{dom}(B^{1/2}) \cap \mathfrak{h} \text{ is dense in } \mathfrak{h}.$$

Let C be the self-adjoint operator on $\mathfrak{h}$ associated with $\mathfrak{k}(h, k) = (B^{1/2}h, B^{1/2}k)$. In its introductory survey, arXiv:0907.1199 proves only time-averaged convergence and states on PDF page 5 that the convergence question in the strong topology remains open:

$$\left(P e^{-itB/n} P \right)^n h \xrightarrow{?} e^{-itC} h \quad (h \in \mathfrak{h}).$$

The particular admissible function at issue is $\phi(x) = e^{-ix}$ [1].

2 The later answer and notation match

Theorem 1.1 of arXiv:2012.15061 (statement beginning on PDF page 1) says: if $H \geq 0$ is self-adjoint, P is orthogonal, and $H^{1/2}P$ is densely defined, then

$$\lim_{n \rightarrow \infty} \left(P e^{-\varepsilon it H/n} P \right)^n f = e^{-\varepsilon it H_P} P f, \quad H_P = (H^{1/2}P)^*(H^{1/2}P), \quad \varepsilon = \pm 1,$$

in Hilbert-space norm for every $f \in \mathcal{H}$, uniformly for t in every bounded real interval [2].

Set $H = B$. Since

$$\text{dom}(H^{1/2}P) = \mathfrak{h}^\perp \oplus (\text{dom}(B^{1/2}) \cap \mathfrak{h}),$$

the later density hypothesis is exactly the source hypothesis. The closed form of H_P is $f \mapsto \|B^{1/2}P f\|^2$, so under $\mathcal{H} = \mathfrak{h} \oplus \mathfrak{h}^\perp$ one has $H_P = C \oplus 0$. Taking $f = h \in \mathfrak{h}$ and $\varepsilon = 1$ therefore gives exactly

$$\left(P e^{-itB/n} P \right)^n h \longrightarrow e^{-itC} h$$

strongly, with the stronger uniformity in bounded t -intervals.

The later authors explicitly describe strong-operator convergence as the question left after their earlier L_{loc}^2 result and state that answering it is the aim of the paper. This is therefore a known later resolution, not a new proof produced by this run.

References

- [1] P. Exner and H. Neidhardt, *Trotter–Kato product formula for unitary groups*, arXiv:0907.1199 (2009), introductory Zeno question on PDF page 5; published as *Remarks on the Trotter–Kato product formula for unitary groups*, Integral Equations Operator Theory 69 (2011), 451–478.
- [2] P. Exner and T. Ichinose, *Note on a product formula related to quantum Zeno dynamics*, arXiv:2012.15061 (2020), Theorem 1.1; Ann. Henri Poincaré 22 (2021), 1669–1697.